Pricing wildfire smoke avoidance: Activity reallocation revealed from mobility data

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Abstract

This paper estimates the welfare cost of avoiding wildfire smoke during the 2018 Northern California fire season. Using individual smartphone location data, I construct ZIP-to-ZIP trip flows by time of day across activity types, including outdoor recreation, work, and other trips. I embed these flows in a nested logit model of activity participation and destination choice, with destination choice estimated as a gravity equation over a comprehensive choice set. Smoke deters both leaving home and visiting polluted locations. I find that people substitute away from smoky outdoor destinations, and that wildfire smoke shifts the composition of daily activities: the share of outdoor activities falls, but so do the shares of work and other trips. The welfare cost from these revealed location choices is around $500M, adding 24% to the mortality burden of the same smoke. Air quality alerts increase avoidance, and under alert-level awareness of moderate smoke, welfare costs would be 20% higher.

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